

Tanishk Yadav

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EDUCATION

New York University, Tandon School of Engineering

Brooklyn, NY

Master of Science in Financial Engineering

05/27

- Coursework: Introduction to Derivative Securities, Quantitative Methods, Credit Risk and Financial Risk Management, Econometrics and Time Series Analysis, Real Time Risk Management, Algorithmic Trading and High Frequency Finance, Corporate Valuation

SRM University (100% scholarship, GPA 3.8/4.0)

Amaravati, India

B.Tech in Computer Science & Engineering (AI & Machine Learning)

06/25

- Coursework: Single/Multi Variable Calculus, Differential Equations, Linear Algebra, Probability, Statistics, Economics

TECHNICAL SKILLS

- **Languages:** Python, C++, C, SQL, R, JavaScript
- **Libraries & Frameworks:** Pandas, NumPy, SciPy, Statsmodels, Scikit-Learn, TensorFlow/Keras, XGBoost, FastAPI, pytest
- **Financial Concepts:** Options Pricing (Black-Scholes, Greeks), Monte Carlo Simulation, VaR/CVaR, Regime Switching (HMM, GMM), Tail Risk Hedging, Bootstrap Resampling, Time Series (ARIMA, AR), Mean-Variance Optimization, Walk-Forward Validation
- **Tools & Platforms:** Bloomberg Terminal, Git, LaTeX, Jupyter, Excel, VS Code
- **Certifications:** Bloomberg Market Concepts (BMC), Bloomberg Finance Fundamentals, Akuna Capital Options 101, Financial Markets (HONORS), AI for Investments, IBM Machine Learning

EXPERIENCE

IUDX (India Urban Data Exchange), Bengaluru, India

06/23 – 08/23

Summer Intern, Framework Development

- Developed a secure, cross-platform digital consent collector using SHA-256 cryptographic hashing, ensuring regulatory compliance for critical data aggregation on the ADeX platform, a government-academia joint venture
- Engineered full-stack JavaScript solutions for consent collection and retrieval as part of a 4-person team, ensuring 100% compliance with central govt. regulations and enabling creation of 38 critical datasets for the government

PROJECTS

ChronoFund: Point-in-Time Fundamental Data Engine | *Python, SEC EDGAR, XBRL, pytest*

- Architected a production-grade data pipeline enforcing zero lookahead bias at **4 independent layers** (FilingsIndex, FilingSelector, XBRLParser, BloombergMapper), each gated by `acceptance_datetime` assertions for second-level PIT precision; **65 unit tests** across cutoff logic, CIK mapping, schema validation, and Bloomberg parsers
- Engineered XBRL parsing with priority-ordered GAAP tag mapping, Parquet output with typed schemas, balance sheet identity checks, survivorship-bias-aware universe tracking via CoverageReport with NaN propagation (no forward-fill), and token-bucket SEC rate limiting (10 RPS)

Regime-Aware Fundamental Strategy | *Python, HMM, Fama-French*

- Built a walk-forward HMM equity strategy (2008–2026) delivering **763% total return (633% net of 2/25 fees)**, **8.37% annualized FF5 alpha** ($t = 2.76$, $p = 0.006$), and 2.45 pp max drawdown reduction vs. SPY through 4-state regime-conditional position sizing
- Implemented composite ranking on 4 fundamental factors (ROIC, leverage, earnings stability, FCF yield) with 4-period EWMA smoothing; applied -30% forced-liquidation penalty after 20 consecutive NaN-price days to eliminate survivorship bias from delistings

VolEdge: Volatility Trading System | *Python, FastAPI, React, Black-Scholes*

- Built a full-stack quantitative trading platform combining Gaussian Mixture Model price distributions with options volatility analysis; computes all Greeks and implied volatility locally via Black-Scholes (Brent root-finding) with no vendor dependency
- Engineered automated trade signal generation from volatility risk premium (VRP) analysis, IV surface construction, term structure, and skew modeling; architected a walk-forward strategy engine with AST-validated sandboxed execution and strict $t-1$ data boundaries

Regime-Adaptive Sector Portfolio Management | *Python, Scikit-Learn, Walk-Forward*

- Designed a multi-scale regime detection system with 4 orthogonal stress detectors (CUSUM sequential analysis, rolling pairwise correlation, market breadth, Fisher skewness) using fuzzy aggregation to avoid binary threshold artifacts
- Achieved **Sharpe 1.08**, **max drawdown -5.25%** vs. SPY's -41.71% (**87% reduction**), and Calmar 1.20 vs. 0.28 over walk-forward validation (2009–2025); all detector thresholds calibrated from training data only